AWAREWAYS

Data-driven PHP developer

Awareways

Technical Assignment: Data Integration Pipeline

The Scenario

Our core platform emits a continuous stream of user activity events. Currently, this data is dumped into raw JSON logs.

The Product team wants to build a new feature: a **"User Activity Dashboard"** that shows users a clean summary of their recent actions and engagement metrics. To do this, they need a reliable, structured way to query this data.

The Challenge

Create an application that records user activity and exposes the activity in a way you see fit. Keep in mind that activity from the platform should eventually stream continuously and be visualised in a graph or dashboard.

Your goal is to bridge the gap between the raw data recorded in this application and the final representation that the Product team can consume.

How you do this is entirely up to you. You decide the architecture, the data model, how to handle malformed records, the framework (or lack thereof), and what the final representation looks like.

Sample user activities:

- User started training
- User made progress
- User scored points
- User completed training

Core Constraints

1. **Timebox:** You should not spend more than just a few hours on this. We want to see your foundational approach, not a feature-complete enterprise application.
2. **Tech Stack:** PHP 8+. You may use any framework or vanilla PHP.

Deliverables

Please submit a Git repository link (or a zipped project) containing:

1. The Code & Setup

- Your source code and whatever database setup/migrations you choose to use.
    - *Tip: To save time, lightweight databases like SQLite are perfectly acceptable.*

2. The README Use the README to explain your engineering decisions. Please include:

- **Instructions:** How to run your application.
- **Usage Examples:** Concrete request/response examples (or CLI examples) so reviewers can quickly understand how to interact with the application without guessing.
- **Project Structure Overview:** A short explanation of how the codebase is organized and where to find the core logic.

3. Be prepared to discuss

- **The "Why":** Why did you choose this specific data model and database? How did you decide to handle the invalid/dirty data?
- **Trade-offs:** What corners did you cut to fit the assignment in a reasonable time?
- **Production Vision:** If this application needed to process 5 million events a day continuously in production, how would your architecture and infrastructure change?